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Commentary

Applying Normal Accident Theory to radiation oncology: Failures are normal but patient harm can be prevented

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There is growing recognition that patient safety is an urgent health care systems problem.[1-4] As we in health care struggle to address this challenge, we might do well to consider the substantial work already done in non–health care settings to better understand and mitigate failures. Experts outside of health care largely embrace the Normal Accident Theory, initially proposed by Dr Charles Perrow in 1984, which provides a set of concepts and framework for analyzing failure potential within and between systems.[5] Normal Accident Theory promotes the concept that failures in systems will occur and should be expected as part of *normal* operations. Systems are defined based on two key characteristics (Fig 1). First, systems in which failures propagate and interact predictably are considered *linear*; those in which failures behave unpredictably are *interactively complex* (Fig 1, x-axis). Second, systems that can adequately detect and respond to failures are *loosely coupled*, whereas systems that cannot detect and respond to failures are *tightly coupled* (Fig 1, y-axis). Examples are shown in Fig 1A.

In this issue of *Practical Radiation Oncology*, Dr Gao and colleagues from the University of Washington report an overall "near-miss event rate" (captured in an in-house incident learning/reporting system) of 0.37 per treatment in the emergent setting versus 0.86 per treatment in the nonemergent setting ($P < .01$). These differences are difficult to interpret because of the potential biases in reporting. The rate of (potentially) severe near misses was substantially higher with emergent treatments, particularly those initiating therapy during weekends/holidays. Because reporting biases are likely lower for severe errors, this finding is likely real and concerning.

How do we interpret these findings through the lens of the Normal Accident Theory? We have attempted to categorize several radiation oncology processes based on their degree of complexity and coupling as per the Normal Accident Theory (Fig 1B).[4] As shown, we believe that some radiation oncology processes are inherently more interactively complex (eg, due to a larger number of participants or variables "in play") and/or more tightly coupled (eg, due to challenges in effectively detecting some failures downstream). Emergent treatments, the focus of the report by Gao et al, might be considered somewhat less interactively complex than other treatments because there are typically a limited number of common "simple" ailments treated emergently (eg, spinal cord or

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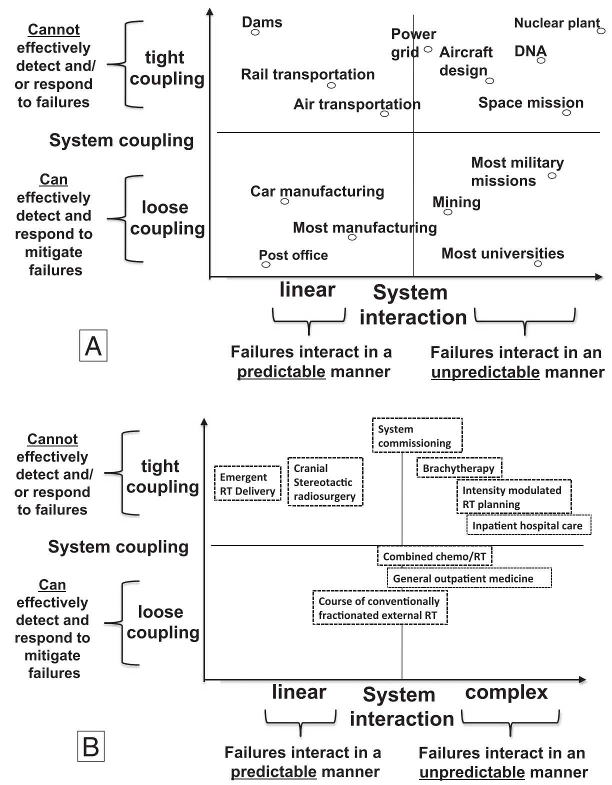

Figure 1 Normal Accident Theory concepts of loose versus tight coupling (y-axis) and linear versus interactively complex. (A) Several nonmedical systems from Perrow's original description are plotted according to Normal Accident Theory concepts (adapted from Perrow[5]). Car manufacturing processes (assembly lines) are considered as linear and loosely coupled systems. Things tend to happen in an orderly fashion (linear), and failures can often be detected and corrected in a timely fashion. Water dams are thought of as linear and tightly coupled systems. A failure at one dam (eg, a breech) typically has predictable downstream (quite literally) consequences (linear). The system is tightly coupled because this is almost certainly going to be manifest (eg, cause a flood). Even though the breech is readily apparent, it is usually not humanly possible to prevent the flood (the time scale for corrective action is much slower than the flow of the water). Universities are seen as complex and loosely coupled systems. Decisions in board areas of the university often affect each other (interactively complex). The unforeseen impacts of these complex interactions are often detected and can be addressed and averted in a timely fashion (loosely coupled). Nuclear power plants are examples of interactively complex and tightly coupled systems. The interactions between the various components are hard to predict, and things can happen very fast—much faster than the response time of a human monitoring the system (tightly coupled). (B) A map of several radiation oncology processes to approximate locations on the Normal Accident Theory grid (adapted from Marks et al[4]).

brain metastases), and thus there might be a limited number of processes to consider and variables "in play." Further, there are usually fewer individuals involved, reducing hand-offs and interactive complexity. On the other hand, emergent after-hours treatments are often done in compacted timelines using nonstandard (or nonroutine) processes that are predisposed to workarounds and cognitive biases during decision making that increase the probability of failures propagating and going undetected (ie, tight coupling). Our industry has not fully engineered (or automated) quality assurance procedures that can be used during emergent treatments (and not all quality assurance can be automated).

Thus, within the Normal Accident Theory construct, we propose that emergent treatments are typically linear and somewhat tightly coupled systems (Fig 1B). Accordingly, such systems can be enhanced by process standardization that includes steps that decrease coupling. Pretreatment chart checks and strategic placement of additional safety checks and timeouts during the emergent treatment process are good examples of this approach. Indeed, Gao et al observed that such pretreatment checks and timeouts caught most of the near misses during emergent treatments. Similarly, physicians and dosimetrists at University of North Carolina engage in a verbal timeout before signing all radiation prescriptions in the treatment planning/record verify systems. This includes an "out loud" verbal verification of treatment site, dose per fraction, number of fractions, total dose, and pacemaker and prior radiation status. This is accompanied by an electronic checklist in our record verify system to attempt to track adherence.

Another strategy to reduce system failure is to dedicate (or "isolate") systems to perform specific functions (ie, to reduce unforeseen interactive complexity) and to add redundancy as needed.[5] At University of North Carolina, we have separately defined emergent after-hours treatment policies and procedures to try to prevent complexity in this setting. Generally, computed tomography simulations are not performed and patients are set up on the treatment machine with rectangular anteroposterior/posteroanterior fields using the onboard image and collimator jaws (no customized blocks). Dose calculations are performed using a simplified in-house treatment planning system. Gao et al reported that the majority of the near misses originated or started in the treatment planning information transfer process. At University of North Carolina, we perform 2 independent emergent treatment planning chart checks, one by the physician and another by the physicist, verifying correct treatment plan information transfer from the treatment planning system to the record verify system.

The strategies described (to reduce coupling and reduce the possibility for unforeseen complexity) will help but will not obviate the problems associated with emergent treatments. During emergent treatments, personnel are limited, team members may not be familiar with each other, and they are asked to work in an environment with heightened stress—adding potential layers of complexity and coupling that are not typically present during routine business hour treatments. Certainly, slowing the system down with verbal time outs and chart checks is useful, but we cannot fully engineer all risks out of emergent treatments. Failures will occur despite our best efforts. Thus, a global strategy to promote safety is needed and should include leadership-inspired safety mindfulness among all workers.

Safety mindfulness is best described as a persistent focus on safety, a recognition that unforeseen events will occur, and a desire to proactively improve their systems and processes.[3,4] We need to create culture, environment, and infrastructure to allow staff and patients to develop an understanding of safety mindfulness and to feel comfortable openly speaking about errors and suboptimal systems. Toward this goal, leaders at University of North Carolina speak often and openly about safety concerns and encourage/recognize and publically celebrate people who participate in improvement work (eg, reporting near misses). Perrow's Normal Accident Theory predicts that accidents are unavoidable in complex settings like ours, and therefore we will need people's active participation to continuously study and improve systems to minimize (or eliminate) serious patient harm, while accepting the fact that systems will fail and human will make errors.

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